

Sliding Discrete Linear Canonical Transform

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Abstract—The linear canonical transform (LCT) has been shown to be one of the most powerful tools in signal processing, and in this paper, we propose an adaptive approach for the computation of the discrete LCT (DLCT), termed the sliding discrete linear canonical transform (SDLCT). First, we introduce a scheme for the single-point DLCT, which can effectively calculate a single or a few linear canonical spectra. Second, the SDLCT is proposed based on an iterative algorithm to meet the requirements of online spectral analysis when only a subset of N frequencies are required from an $\tilde{N}$ -point discrete LCT ($N \leq \tilde{N}$). The additivity and reversibility properties of the proposed algorithms are also discussed in detail. Third, the DLCT convolution operation is obtained to reduce the spectral leakage of the proposed algorithm, and time-domain windowing is implemented via frequency-domain convolution. Finally, we present two methods to assess performance with regard to computational complexity and precision and to show the correctness of the derived results.

Index Terms—Linear canonical transform, discrete linear canonical transform, Horner algorithm, sliding discrete linear canonical transform.

I. INTRODUCTION

TRANSFORMS are widely employed in signal processing to acquire useful information that is not explicitly available when the signal is in the time domain. Most real-time signals, such as speech and biomedical signals, are non-stationary signals. Many useful tools have been developed for processing these non-stationary signals [1]–[7]. Among them, the LCT, which is a three-parameter linear integral transform, has been proven to be one of the most powerful tools. The LCT of a signal $x(t)$ with real parameters α , β and γ is defined as [2], [3]

$$X^A[x(t)](u) = \int_{-\infty}^{\infty} x(t) K_A(u, t) dt,$$

where $K_A(u, t) = \sqrt{\beta} e^{-\frac{j\pi}{4}} e^{j\pi(\alpha u^2 - 2\beta ut + \gamma t^2)}$ is the LCT transform kernel, $A = [\gamma/\beta, 1/\beta; -\beta + \alpha\gamma/\beta, \alpha/\beta] \in SL(2, R)$. The LCT operator is unitary, $K_{A^{-1}}(t, u) = K_A^*(u, t)$. Thus, the inverse LCT can be obtained by the LCT with parameter matrix

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A^{-1} . It has been shown that the LCT has many applications in the optics and signal processing community [8]–[10], and it has recently received substantial research interest [6]–[18]. These research results indicate that the LCT has gradually become a powerful signal processing tool.

The discretization method and a fast algorithm are one of the keys to the application of a transformation. Therefore, research on discrete methods for the LCT has become a hot research topic in recent years. In the next section, we first review the relevant research results of DLCT, and then, we present the research ideas and contributions of this article.

A. Review of Related Works

The simplest way to derive the DLCT is by directly sampling the continuous LCT in both the time domain and the LCT domain. To use the DLCT to approximate the samples of the LCT of a continuous signal very accurately, it is necessary to know how to choose the number of samples and the sample interval. This issue is answered thoroughly in chapter 7 of the book [7]. The authors of [7] discussed the bicanonical width product, which determines the number of sampling points. This is important for samples and effective numerical calculations.

In [19], a discrete LCT (DLCT) was first introduced by Pei and Ding in 2000. This DLCT can be effectively computed by successively performing a chirp multiplication, a fast Fourier transform (FFT) and a second chirp multiplication. Later, a split-radix method that decomposed the discrete transform matrix into smaller matrices iteratively was first utilized by Hennesly and Sheridan to quickly implement the DLCT [20]–[22]. In addition, benefiting from the additivity property, the LCT operator can be decomposed into products of special operators [23]–[28]. In 2006, Ozaktas *et al.* proposed a digital algorithm based on Iwasawa decomposition for the LCTs in [24], which has the same computational complexity and calculation speed as the FFT. In [27], Koç and Ozaktas analyzed a variety of decompositions of the LCT, and they proposed two digital computation algorithms. Their methods, associated with oversampling operations, focused on the adequacy of sampling such that the continuous LCT could be recovered from its samples. In chapter 8 of the book [7], Healy and Ozaktas explored the relationship of the LCTs with sampling, and they studied how to achieve sampling rate changes accurately and efficiently, as well as provided a new understanding of DLCTs.

In [28], the exact relationship, in terms of the interval dependence, between the DLCT and continuous LCT was derived,

being expressed as [28]

$$X^A(m) = C \sum_{k=-N/2}^{N/2-1} K_A(m, k)x(k), \quad (1)$$

where $K_A(m, k) = e^{\frac{j\pi}{N|\beta|}(\alpha m^2 - 2\beta km + \gamma k^2)}$ is the kernel of the DLCT and $C = \frac{\sqrt{\beta}e^{-\frac{j\pi}{4}}}{\sqrt{N|\beta|}}$. The corresponding inverse transformation of $X^A(m)$ is given by

$$x(k) = C^* \sum_{m=-N/2}^{N/2-1} X^A(m)K_A^*(m, k), \quad (2)$$

where $K_A^*(m, k)$ and C^* are the conjugates of $K_A(m, k)$ and C , respectively. This DLCT has a simple closed form with complexity $O(N \log_2 N)$, and it preserves many properties of the LCT. It is an important candidate for a widely accepted definition of fast DLCT. Combining the transform matrix decomposition with the interval-independent DLCT, Pei *et al.* in [29] proposed another DLCT based on the CM-CC-CM and FFT, which also did not alter the sampling rate of the signal during the algorithm (i.e., no interpolation or decimation). In the same year, Koç and Ozaktas *et al.* analyzed and compared the DLCT with complexity $N \log_2 N$ in chapter 10 of the book [7]. In addition to the above-mentioned algorithms, various other fast discretization methods have been proposed in [30]–[33]. These research results provide a good basis for the further development of the DLCT toward meeting the requirements of practical applications. Many aspects of the fast methods of the DLCT still need to be studied.

B. Contributions of This Paper

To the best of our knowledge, the above discrete algorithms use a block of N input samples to calculate all N output spectra. These fast implementation algorithms can improve the computational complexity and efficiency for very large N . However, for certain applications, we only need to calculate a sample of the output spectral component or the DLCT of the stream data, which is updated with every new sample. In these cases, the direct application of the fast DLCT may result in excessive computational costs for every N input samples.

Thus, to overcome the aforementioned problems and achieve a unique or several LCT spectrum bins, we propose a sliding discrete LCT in this paper. First, the single-point DLCT based on the Horner algorithm [34], which can effectively calculate a single or a few samples of output spectral components, is derived. Furthermore, for real-time signal processing, the stream data of the input signals arrive in a sequential manner. The transform of the signal is obtained by sequentially processing the blocks of N samples. When there is a maximum repeatability between adjacent blocks of data, that is, the block of N samples is updated with every new sample, the other type of DLCT, called the sliding DLCT (SDLCT), which uses the spectra of the previous block to iteratively compute the spectra of the next block, is proposed. This sliding technique has been successfully employed in computing the discrete Fourier transform (DFT) and is called the sliding discrete Fourier transform (SDFT) [35],

which is our special case. Finally, to reduce the spectral leakage of the SDLCT, the DLCT convolution theorem is presented and the process of frequency-domain convolution for accomplishing time-domain windowing is obtained.

This paper is organized as follows. In Section II, the Sp-DLCT and SDLCT are derived for implementing the calculation of a single sample or a few samples of the discrete transform. The properties and computational complexity of the proposed algorithms are investigated. In Section III, the DLCT convolution theorem is presented and the calculation of the DLCTs of data with a non-rectangular window is obtained. The performances of the proposed methods are presented and discussed in Section IV. Finally, conclusions are drawn in Section V.

II. SINGLE-POINT DLCT AND SLIDING DLCT

A. Single-Point DLCT Algorithm (Sp-DLCT)

To obtain a single spectrum at m_0 , we propose the Sp-DLCT based on the Horner algorithm [34]. First, we convert the definition of the DLCT (1) into the following form,

$$X^A(m_0) = C e^{j\pi \frac{\alpha m_0^2}{N|\beta|}} \sum_{k=-N/2}^{N/2-1} b_k z^k = C e^{j\pi \frac{\alpha m_0^2}{N|\beta|}} B(z), \quad (3)$$

where $k = -N/2, \dots, N/2 - 1$

$$B(z) = \sum_{k=-N/2}^{N/2-1} b_k z^k, \quad (4)$$

$$b_k = x(k) e^{j\pi \frac{\gamma k^2}{N|\beta|}}, z = e^{j\pi \frac{1}{N|\beta|}(-2\beta m_0)}. \quad (5)$$

Note that the sequence b_k is dependent on β and γ . For different parameters β and γ , we therefore need to recalculate it. This makes the algorithm inflexible. Fortunately, the following interesting recursive method provides a convenient solution for addressing these disadvantages.

Lemma 1: Let $D_k = e^{j\pi \gamma (2k+1)/N|\beta|}$ and $g_k = e^{j\pi \gamma k^2/N|\beta|}$, $k = -N/2, \dots, N/2 - 1$; then, D_k and g_k have the following recurrence relations:

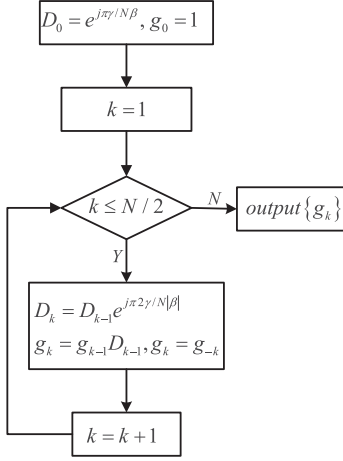
$$D_{k+1} = e^{j\pi \frac{2\gamma}{N|\beta|}} D_k, \quad g_{k+1} = g_k D_k, \quad (6)$$

and $D_0 = e^{j\pi \frac{\gamma}{N|\beta|}}$, $g_0 = 1$.

We omit the proof of Lemma 1 here because it is easy to verify by mathematical derivation. It is noted that the sequence g_k is an even sequence, so we only need to calculate g_k for $0 \leq k \leq N/2$ and use $g_k = g_{-k}$ to obtain the entire sequence. Fig. 1 is the implementation structure of g_k . The recursive process only requires D_0 and g_0 to obtain g_k , thereby avoiding recalculation of the power series for different parameters. Then, we utilize g_k to obtain b_k in (5).

Finally, if $m_0 \neq 0$, the summation in (4) can be obtained by the following recursive algorithm:

$$\begin{aligned} B(z) &= 0, \\ \text{for } i &= -N/2, -N/2 + 1, \dots, N/2 - 1, \\ B(z) &= b_i + B(z)z. \end{aligned} \quad (7)$$

Fig. 1. Implementation structure of g_k .

Thus, we can obtain $B(z)$ for $i = N/2 - 1$. The resulting $B(z)$ is multiplied by $Ce^{j\pi\alpha m_0^2/(N|\beta|)}$, thus, a specific LCT spectra can be obtained.

If $m_0 = 0$, equation (1) is rewritten as

$$X^A(0) = C \sum_{k=-N/2}^{N/2-1} x(k)g_k, \quad (8)$$

we only need to use the obtained g_k direct summation (8).

The above discussion provides a general calculation method for a specific LCT spectra, which we call the single-point DLCT (Sp-DLCT). Fig. 2 shows the implementation of the Sp-DLCT. The advantages of the Sp-DLCT are that it reduces the operation of complex exponentiation and reduces the computational storage requirements. The complexity of the Sp-DLCT is N complex multiplications and N complex additions, which is the same as the direct use of the DLCT (1). However, the direct algorithm entails storing approximately N times the number of bits of z (the evaluated $B(z)$ has an approximate magnitude $z^{N/2}$, and one must also store $z^{N/2}$ itself). By contrast, the Sp-DLCT's storage requirement is only $N/2$ times the number of bits of z . Therefore, the Sp-DLCT requires less storage than the direct use of the DLCT (1).

For real-time signal processing, the transform of a signal is obtained by sequentially processing the blocks of N samples. This is equivalent to adding a rectangular window with length N to the input signal. Then, the block of N samples is updated, which is similar to shifting the window over time. As long as the data is updated, the Sp-DLCT needs to recalculate the output spectra. In this processing, the similarity among blocks of samples is not considered. For this scenario, we propose a novel type of recursive algorithm, which we name the SDLCT, to implement the continuous spectra analysis.

B. Sliding DLCT Algorithm (SDLCT)

When the window slides along one sample at a time, the transform can be computed by processing a new block of samples consisting of the newly arrived sample and the $N - 1$ samples of the previous block. This is referred to as the sliding technique. Fig. 3, for example, shows the sliding algorithm performing an

N -point DLCT within a sliding window. In this example, the SDLCT initially computes the DLCT of $N = 20$ samples in Fig. 3(a). The time window is then advanced one sample, as in Fig. 3(b), and a new N -point DLCT is calculated.

We introduce the time-domain index n for the continuous calculation spectra. Then, we denote the N samples within the sliding window as $x(n-1, k) = x[n-1-(N/2-1)+k]$, $k = -N/2, \dots, N/2-1$ at the $(n-1)$ th moment. We utilize the chirp periodic, time-shift and frequency-shift properties of the DLCT [20] to obtain the expressions for the SDLCT. This type of SDLCT admits a recursive representation as stated in the following theorem.

Theorem 1: The DLCT of the windowing data $x(n-1, k)$ with parameter matrix $A = [\gamma/\beta, 1/\beta; -\beta + \alpha\gamma/\beta, \alpha/\beta]$ is $X_{n-1}^A(m)$, and the DLCT of $x(n, k)$ is $X_n^A(m)$. Then, we have the following recursive relations between $X_{n-1}^A(m)$ and $X_n^A(m)$,

When γ/β is an integer,

$$\begin{aligned} X_n^A(m) = & e^{j\pi \frac{1}{N|\beta|} [-\alpha(2m\frac{\gamma}{\beta} + \frac{\gamma^2}{\beta^2}) + 2m\beta + \gamma]} X_{n-1}^A\left(m + \frac{\gamma}{\beta}\right) \\ & - Cx(n-N)K_A\left(m, \frac{-N}{2} - 1\right) \\ & + Cx(n)K_A\left(m, \frac{N}{2} - 1\right), \end{aligned} \quad (9)$$

When γ/β is not an integer,

$$\begin{aligned} X_n^A(m) = & e^{j\pi \frac{1}{N|\beta|} (2\beta m + \gamma)} G_{n-1}^A(m) - Cx(n-N) \\ & \times K_A\left(m, \frac{-N}{2} - 1\right) + Cx(n)K_A\left(m, \frac{N}{2} - 1\right) \end{aligned} \quad (10)$$

where N is the sliding window length, $X_n^A(m)$ is the new spectral component and $X_{n-1}^A(m + \gamma/\beta)$ is the previous spectral component. m denotes spectra associated with the m th DLCT bin. $G_{n-1}^A(m)$ is the m th DLCT of the $(n-1)$ th of the signal $g(n-1, k) = x[n-1-(N/2-1)+k]e^{-j2\pi\gamma k/(N|\beta|)}$.

Proof: See Appendix A. ■

Remark 1: It should be noted that as time progresses, the window length is fixed for the given signal and selected transform parameters. If the input signals are changed, N may be changed accordingly. The length of the window is determined by the prior information of the signal and the actual requirements such as avoiding aliasing of the signal in the time domain and the frequency interval to be observed.

Fig. 4 shows the implementation structure of the SDLCT. One of the attributes of the SDLCT is that once we have obtained the spectra of the previous moment $X_{n-1}^A(m + \gamma/\beta)$, we can utilize (9) or (10) to obtain the spectra of the next moment. If the window slides continuously, the continuous spectra analysis can be implemented for the specified spectrum bin. When we choose the different m , we can obtain different output spectra bins. If we select the $m = -N/2, \dots, N/2-1$, all the spectrum bins can be obtained. Therefore, the SDLCT can be used to compute a single spectrum bin, all N spectrum bins or the

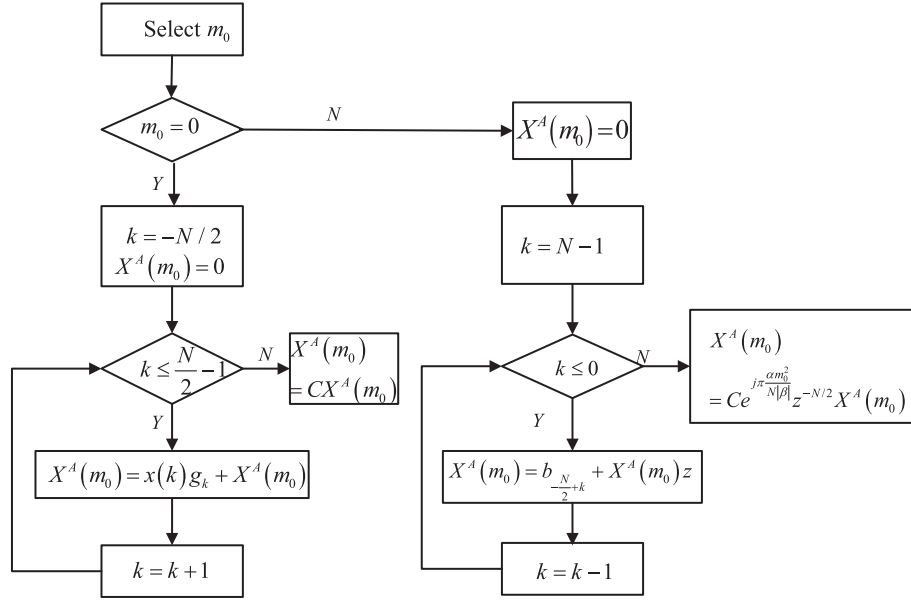
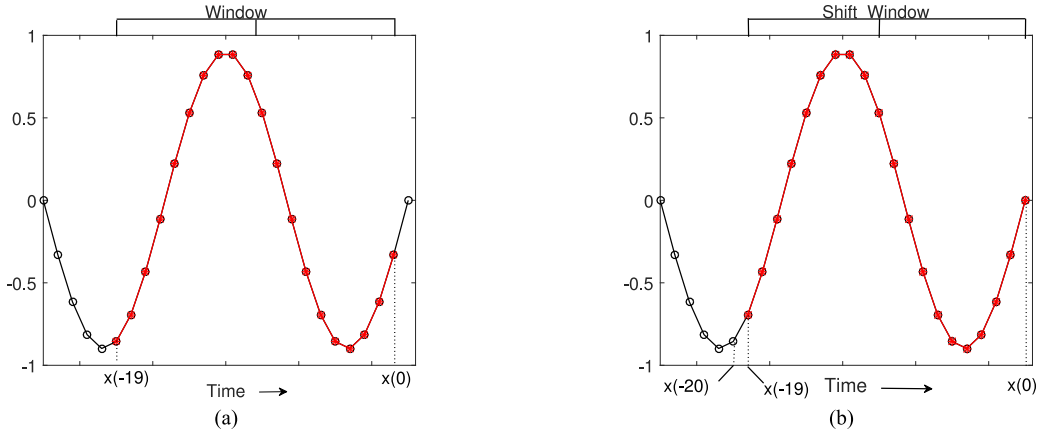
Fig. 2. Implementation structure of $X^A(m_0)$.

Fig. 3. Signal windowing for two 20-point DLCTs, (a) data samples in the first computation, (b) second-computation samples.

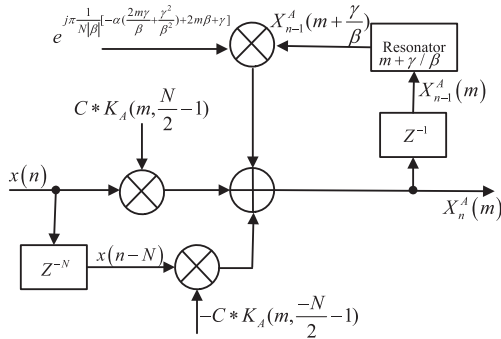


Fig. 4. Implementation structure of SDLCT.

desired frequency range according to the given application requirements. Based on the above facts, the proposed SDLCT can realize the selectivity of the spectra output and the performance of time-frequency analysis.

When α, β, γ are special values, we obtain the sliding discrete transform of the corresponding special cases. The SDFT proposed in [35] is one of our special cases when $\alpha = 0, \beta = 1$ and $\gamma = 0$. The properties of the Sp-DLCT and SDLCT will be discussed in the next section.

C. Additivity and Reversibility

The LCT has two important properties: additivity and reversibility. The additivity property implies that several LCTs that are cascade operated are equivalent to a single combined LCT. The reversibility property provides an inversion of the LCT, that is, the inverse LCT. The ability to recover original signals is guaranteed by this reversibility. However, the properties of the discrete transforms are very different from those of continuous cases. It was shown in earlier papers [21], [22], [24], [27] that the discretization may destroy the additivity and unitary properties of the continuous LCT because the discrete transform

TABLE I
NMSES OF ADDITIVITY PROPERTY BETWEEN IDLCT SIGNALS AND ORIGINAL SIGNALS BY (12)

	$x_1/(A_2A_1)$	$x_1/(A_4A_3)$	$x_2/(A_2A_1)$	$x_2/(A_4A_3)$
Sp-DLCT	1.1×10^{-3}	1.4×10^{-4}	1.88×10^{-2}	3.8×10^{-3}
SDLCT	7.4×10^{-3}	5.6×10^{-3}	4.33×10^{-2}	3.5×10^{-3}

is obtained by sampling the continuous LCT kernels. Thus, we discuss properties of the Sp-DLCT and SDLCT in this section.

1) *Additivity*: The additivity property for the DLCTs is defined as follows:

$$X_n^{A_1} X_n^{A_2} = X_n^{A_1 \times A_2}. \quad (11)$$

Equation (11) reveals that using the proposed methods, successively applying two DLCTs is equivalent to applying a single DLCT. The additivity property (11) can be further extended to three or more DLCT compositions. In the following, we examine the normalized mean-square error (NMSE) of the additivity property (11) for the proposed methods:

$$\text{NMSE} = \frac{\sum_m |X_n^{A_2}(X_n^{A_1}x_i(n, k)) - X_n^{A_2 \times A_1}(x_i(n, k))|^2}{\sum_m |X_n^{A_2 \times A_1}(x_i(n, k))|^2}. \quad (12)$$

Here, the discrete signals are described as follows:

$$x_1(k) = e^{-k^2 \pi / N},$$

$$x_2(k) = e^{-k^2 / N} [2 \cos(2\pi k / \sqrt{N}) - j \sin(\pi k / \sqrt{N} - 1)].$$

Again, we take the parameter matrices as follows: $A_1 = [1/2, 1/2; -8/5, 2/5]$; $A_2 = [2/5, 1/3; -11/5, 2/3]$; $A_3 = [5/11, 5/11; 237/110, 5/22]$; $A_4 = [1/2, 1/2; -3/2, 1/2]$. The input data length $\tilde{N} = 128$, and the window length $N = 126$. Here, the NMSEs in (12) of $x_1(k)$, with the LCT matrix A_1 followed by A_2 and A_3 followed by A_4 , are denoted $x_1/(A_2A_1)$ and $x_1/(A_4A_3)$, respectively. We also obtain $x_2/(A_2A_1)$ and $x_2/(A_4A_3)$ for $x_2(k)$ with a similar method. The simulation results are shown in Table I. Note that the NMSEs of the Sp-DLCT and SDLCT are almost identical for different signals and parameters.

2) *Reversibility*: The reversibility property of the proposed algorithms is examined through the following NMSE method:

$$\text{NMSE} = \frac{\sum_k |x_i(n, k) - X_n^{A^{-1}} X_n^A x_i(n, k)|^2}{\sum_k |x_i(n, k)|^2}, \quad (13)$$

The following three types of signals are used in the simulations:

$$x_1(k) = \cos(\pi k^2 + \pi k) + 10j \sin(\pi k^2 + \pi k);$$

$$x_2(k) = 30e^{-10j\pi k^2 / N}; \quad x_3(k) = \text{rect}(k/N);$$

where the sequence length $\tilde{N} = 128$, the window length $N = 100$ and $k = -N/2, \dots, N/2 - 1$. Let the parameter matrix be $A = [2, 1/2; -9/5, 1/20]$, and the NMSEs between the inverse DLCT of signals and the original signals are listed in Table II. The table shows that all the NMSEs of the Sp-DLCT are less

TABLE II
NMSES OF REVERSIBILITY PROPERTY BETWEEN IDLCT SIGNALS AND ORIGINAL SIGNALS BY (13)

	x_1	x_2	x_3
Sp-DLCT	4.11×10^{-29}	7.62×10^{-29}	1.84×10^{-27}
SDLCT	5.52×10^{-7}	1.50×10^{-3}	9.47×10^{-4}

than 10^{-27} . Therefore, the Sp-DLCT is superior to the SDLCT in terms of the reversibility property. Thereby, it is unnecessary to develop the inverse DLCT additionally because it can be realized by the forward DLCT with A^{-1} .

D. Computational Complexity

In this section, we analyze in detail the computations required to perform the Sp-DLCT and SDLCT. The Sp-DLCT involves $4N + 8$ real multiplications and $2N$ real additions to obtain a single LCT spectra, which is associated with a window length N . When the number of output spectra to be observed is M' , it is obvious that the number of real multiplications is $(4N + 8)M'$ and that of real additions is $2NM'$. We can also derive that the Sp-DLCT has a higher computational efficiency than the fast discrete linear canonical transform (FDLCT) [28] in the case of $M' < \min \{ \lceil 2 + (N \log_2 N - 8)/(2N + 4) \rceil, \lceil \log_2 N \rceil \}$. For the SDLCT, once the previous spectra has been obtained, the calculation of the next spectra requires 24 real multiplications and 4 real additions per output sample. Table III shows comparisons of the computational complexity for computing a single sample of output components.

For the sliding window length, N , we calculate a single DLCT spectra at M succession moments. The computations are $(4N + 8)M$ real multiplications and $2NM$ real additions and $12M(M + 1)$ real multiplications and $2M(M + 1)$ real additions using the Sp-DLCT and SDLCT, respectively. When $M \leq \lfloor \frac{N-1}{3} \rfloor$ (the symbol $\lfloor * \rfloor$ represents a maximum integer no greater than $*$), the performance of the SDLCT is better than that of the Sp-DLCT. When $N = 200$, Fig. 5 shows the complexity of the Sp-DLCT and SDLCT. When $M \leq 66$, the SDLCT requires fewer computations than the Sp-DLCT.

When we calculate the blocks of N samples, all the output spectra for M consecutive different moments, as long as the signal spectra of the previous moment are obtained, the computational complexity of each iteration is only $24N$ real multiplications and $4N$ real additions for the SDLCT. The total computational complexity is $24NM$ real multiplications and $4NM$ real additions for the SDLCT, while the Sp-DLCT and FDLCT need to be calculated separately for each moment. The total computation is $(4N + 8)NM$ real multiplications and $2N^2M$ real additions for the Sp-DLCT and $(8N + 2N \log_2 N)M$ real multiplications and $2MN \log_2 N$ real additions for the FDLCT.

Based on the previous analysis, the Sp-DLCT and SDLCT have an advantage over the FDLCT in terms of computing a single or a few LCT spectra bins. When the window slides a large number of times, the Sp-DLCT is more computationally

TABLE III
COMPARISONS OF SINGLE-POINT DLCT COMPUTATIONAL COMPLEXITY

	Single $X_n(m)$ computation		Next single $X_{n+1}(m)$ computation	
	Real adds	Real multiplications	Real adds	Real multiplications
FDLCT	$2N \log_2 N$	$8N+2N \log_2 N$	$2N \log_2 N$	$8N+2N \log_2 N$
Sp-DLCT	$2N$	$4N + 8$	$2N$	$4N + 8$
SDLCT	$2N \log_2 N$	$8N+2N \log_2 N$	4	24

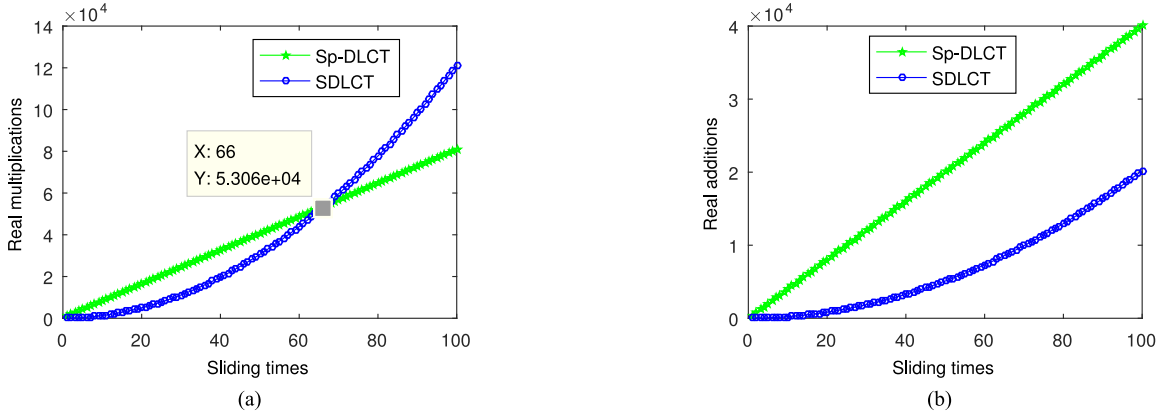


Fig. 5. When a sliding window length N of 200, comparison of (a) real multiplications and (b) real additions between the Sp-DLCT and SDLCT.

efficient than the SDLCT for calculating a single LCT spectra bin. When we calculate all LCT spectra in each window, the SDLCT has a lower computational complexity than the Sp-DLCT.

The SDLCT processes the signal in a limited time, and similar to the DFT, it adds a rectangular window to the signal, in which the sampling signals are considered 0 outside the window and intercepted as N samples in the window. However, the truncation can cause spectral leakage. In the next section, we focus on discussing this problem.

III. TIME-DOMAIN WINDOWING IN THE FREQUENCY DOMAIN

The spectral leakage of the SDLCT can be reduced by adding a non-rectangular window to the input time samples $x(k)$. Time-domain adding non-rectangular window would mean that the computational simplicity of (9) or (10) would not be established. Thus, we can implement time-domain windowing via frequency-domain convolution.

The convolution theorems for the continuous LCT have been derived in [36]–[38]. In this paper, we derive the discrete convolution formula for the LCT based on the convolution theorem for the continuous LCT [36] as follows:

$$\begin{aligned}
 S^A(m) &= X^A[x(k)w(k)](m) \\
 &= |C|^2 e^{\frac{j\pi\alpha m^2}{N|\beta|}} \left[e^{-\frac{j\pi\alpha m^2}{N|\beta|}} X^A(m) * W\left(\frac{\beta}{|\beta|}m\right) \right], \quad (14)
 \end{aligned}$$

where $X^A(m)$ and $W(m)$ are the DLCT and DFT, respectively, of $x(k)$ and the window function $w(k)$, $k, m =$

$-N/2, \dots, N/2 - 1$, and $*$ is the classical convolution operation. Spectral leakage reduction in the frequency domain is accomplished by convolving $X^A(m)$ with the DFT of a non-rectangular window function. Considering the spectra of the Hanning window, the Hamming window and Blackman window, which are members of a category called $\cos^\alpha(t)$ window functions and can be used to reduce spectral leakage in [39], are easy to implement in hardware. Thus, we choose these functions as window functions. The N -point time-domain samples of these window functions are defined as

$$w(k) = \sum_{l=0}^{L-1} (-1)^l \alpha_l \cos(2\pi l k / N), \quad (15)$$

where $k = -N/2, \dots, N/2 - 1$, α_l is the window function coefficient, and the integer L specifies the number of terms in the window's function. The DLCT of a signal with the non-rectangular window function $w(k)$ in (15) can be implemented by the following Theorem 2.

Theorem 2: Suppose that $s(n, k) = x(n, k)w(k)$ is windowed by the n th-time discrete data, $w(k)$ is the window function defined in (15), and $S_n^A(m)$ is the DLCT of $s(n, k)$ with parameter matrix $A = [\gamma/\beta, 1/\beta; -\beta + \alpha\gamma/\beta, \alpha/\beta]$; then, we can compute the DLCT of $s(n, k)$, $S_n^A(m)$, as follows:

$$\begin{aligned}
 S_n^A(m) &= |C|^2 e^{j\pi \frac{\alpha m^2}{N|\beta|}} \frac{1}{2} \sum_{l=0}^{K-1} (-1)^l \alpha_l \\
 &\times \left[e^{-j\pi \frac{\alpha(m+l)^2}{N|\beta|}} X_n^A(m+l) + e^{-j\pi \frac{\alpha(m-l)^2}{N|\beta|}} X_n^A(m-l) \right],
 \end{aligned}$$

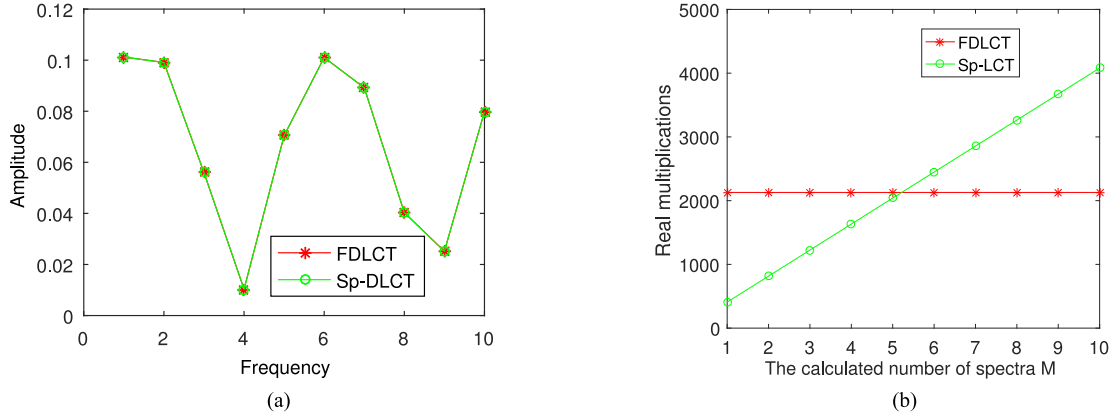


Fig. 6. Simulation results (a) local spectra of the FDLCT and Sp-DLCT in $[1, 10]$, (b) real multiplications of the FDLCT and Sp-DLCT.

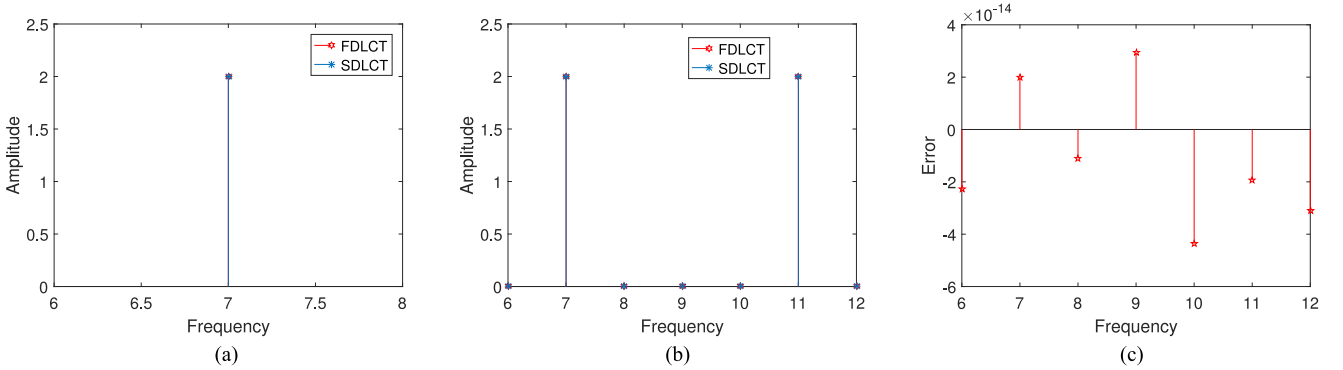


Fig. 7. Amplitude of the FDLCT and SDLCT for the sliding window length, $N = 20$. (a) a single frequency spectra sequence, (b) a part of the frequency spectra sequence, (c) error between FDLCT and SDLCT for calculating the part of LCT frequency.

where $X_n^A(m)$ is the m th DLCT bin of the n th-time discrete data $x(n, k)$.

Proof: See Appendix B. ■

Based on this theorem, we can utilize the convolution of the DFT of the window $W(m)$ with the spectra $X_n^A(m)$ to reduce the spectral leakage. For example, the DFT of a Hanning window consists of only three nonzero values: -0.25 , 0.5 , and -0.25 . We can compute the $S_n^A(m)$ using three-point convolution:

$$S_n^A(m) = \frac{1}{4} |C|^2 e^{j\pi \frac{\alpha m^2}{N|\beta|}} \left[2e^{-j\pi \frac{\alpha m^2}{N|\beta|}} X_n^A(m) - e^{-j\pi \frac{\alpha(m+1)^2}{N|\beta|}} X_n^A(m+1) - e^{-j\pi \frac{\alpha(m-1)^2}{N|\beta|}} X_n^A(m-1) \right].$$

If the output is allowed to have a gain of 4, the structure of the frequency-domain convolution is further simplified as

$$S_n^A(m) = |C|^2 e^{j\pi \frac{\alpha m^2}{N|\beta|}} \left[2e^{-j\pi \frac{\alpha m^2}{N|\beta|}} X_n^A(m) - e^{-j\pi \frac{\alpha(m+1)^2}{N|\beta|}} X_n^A(m+1) - e^{-j\pi \frac{\alpha(m-1)^2}{N|\beta|}} X_n^A(m-1) \right]. \quad (16)$$

From (16), we find that the frequency-domain convolution utilizes adjacent frequency units of the SDLCT, $m-1, m, m+1$, to perform four multiplications and two additions, instead of N multiplications of the window function coefficient with the in-

put signal. This method guarantees the simplicity of the SDLCT and reduces spectral leakage and the number of multiplications.

Remark 2: All the above conclusions are based on the DLCT derived by Ozaktas's algorithm in [28], and for the other type of discrete algorithms proposed in [19], [20], [29], similar conclusions can be obtained by our methods.

IV. SIMULATIONS AND COMPARISONS

To demonstrate the effectiveness and performance of the proposed algorithms, simulations and comparisons with current methods are given in this section.

A. Sp-DLCT Simulations

In this section, we utilize the rectangular window signal $x(k) = \text{rect}(k/10)$, with $-50 \leq k \leq 49$, to verify the correctness and advantages of the proposed Sp-DLCT by comparison with current methods [28].

We take the parameters $\alpha = 2.0260, \beta = 1, \gamma = 0.4243$ and the sliding window length $N = 100$. We calculate the DLCT of $x(k)$ using the Sp-DLCT and existing FDLCT [28] to calculate the local spectra. The corresponding simulation results are shown in Fig. 6(a). The results suggest that the Sp-DLCT has almost the same accuracy as the FDLCT. Fig. 6(b) shows the variation in the number of real multiplications with the number of calculated spectra for the Sp-DLCT and FDLCT. The results

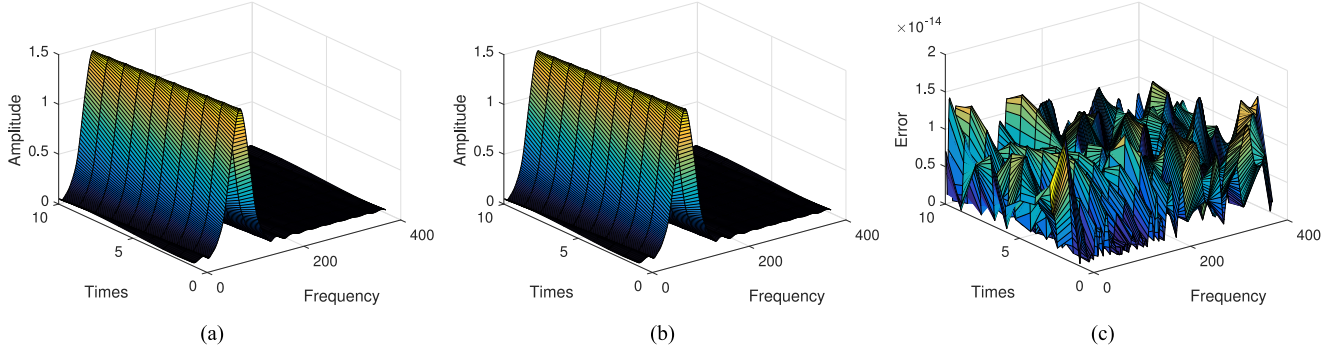


Fig. 8. Amplitude using (a) the SDLCT and (b) FDLCT, (c) error of the amplitude between the FDLCT and SDLCT. The transform parameters are $\alpha = 0.258, \beta = 0.25, \gamma = 0.75$, the sliding times is $M = 10$ for the discrete signal $x(k)$.

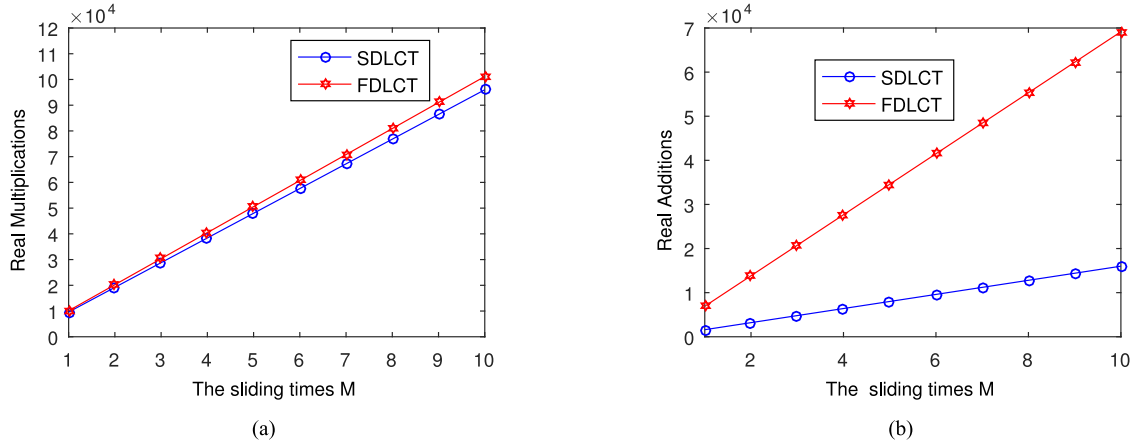


Fig. 9. (a) Real multiplications of the FDLCT and SDLCT; (b) real additions of the FDLCT and SDLCT. The transform parameters are $\alpha = 0.258, \beta = 0.25, \gamma = 0.75$, the sliding window length is $N = 400$.

indicate that the number of calculations is less than five, and the Sp-DLCT has lower complexity than the FDLCT. Therefore, the Sp-DLCT is more effective than the FDLCT in calculating a few LCT spectra without sacrificing accuracy.

B. SDLCT Simulations

1) *Simulations for SDLCT-Calculated Every Few Spectra or a Single Spectrum:* To verify the effectiveness of the theory of the SDLCT, we consider the following signal:

$$x(k) = e^{j20\pi(k/10)^2 + j\pi k}, \quad (17)$$

where $-50 \leq k < 50$. We take the transform parameters as $\alpha = 1.859, \beta = 2$ and $\gamma = 0$, and the length of the sliding window $N = 20$. Next, we analyze a two-consecutive-moment spectra in terms of precision and computational complexity.

Fig. 7 shows a single spectra and the local range spectra. The results suggest that the proposed methods can obtain the same precision as the FDLCT. When the number of successive sliding operations M is varied, Table IV shows a quantitative comparison of the computational complexity using the Sp-DLCT, SDLCT and FDLCT to calculate a single output frequency. The results indicate that in the case of $M < 9$, the SDLCT performs better than the Sp-DLCT, and both the SDLCT and the

TABLE IV
COMPARISON OF THE COMPUTATIONAL REAL MULTIPLICATIONS OF SINGLE-POINT SPECTRA FOR DIFFERENT SLIDING TIMES M WHEN THE SLIDING WINDOW LENGTH IS $N = 20$

M	1	5	9	13	17
FDLCT	333	1664	2996	4327	5659
Sp-DLCT	88	440	792	1144	1496
SDLCT	24	360	1080	2184	3672

Sp-DLCT have lower computational complexities than the FDLCT for calculating a single spectra bin.

2) *Simulations for SDLCT-Calculated Spectra of the Entire Window at Different Times:* In this section, we calculate all the spectra in the entire sliding window at different times using the SDLCT to show the variation in the local LCT frequency over time. To demonstrate the generality of the SDLCT algorithm, we consider the following signal $x(k)$:

$$x(k) = 1.5tri(3k/20) - 0.5tri[(k + 300)k/20], \quad (18)$$

where $tri(\cdot)$ is triangular function [27], [29], and $-300 \leq k < 300$. We take $\alpha = 0.258, \beta = 0.25, \gamma = 0.75$ and a sliding

window length $N = 400$, and we utilize the SDLCT to calculate all the spectra in the window at different times.

In this section, we show the spectra of the first ten times, that is, $M = 10$. We first calculate the first moment DLCT of the windowed data $x(k)$, $-N/2 \leq k < N/2$, by the FDLCT; then, we utilize (9) to iteratively calculate the whole spectra in the sliding window. Figs. 8(a) and 8(b) show the amplitudes of the discrete signal $x(k)$ calculated by the SDLCT and FDLCT, respectively. Using Figs. 8(a) and 8(b), we can visualize the variation of a specified spectra over time, thereby showing the performance of the algorithm's time-frequency analysis. Fig. 8(c) shows the amplitude error between the SDLCT and the FDLCT and evidences the fact that the SDLCT achieves the same accuracy as the FDLCT for the spectra of blocks. Figs. 9(a) and 9(b) show the complexity of the SDLCT and FDLCT, and they indicate that the SDLCT has lower complexity than the FDLCT.

The above simulations show that the theoretical results are consistent with practical applications. The proposed SDLCT with a local window function provides a new tool for the joint analysis of the time domain and LCT domain.

V. CONCLUSIONS

In this paper, we have introduced the SDLCT for some real-time applications. In this process, the window is shifted one sample at a time and repeated continuously. The Sp-DLCT is proposed based on the well-known Horner algorithm, which transforms the complex exponentiation into linear iteration. One advantage of the Sp-DLCT is that it can efficiently calculate a single or a few samples of output spectral components in each window. It is useful for analyzing a particular spectrum bin of signals in practical applications. In addition, the Sp-DLCT saves on storage space because its storage requirement is only $N/2$ times the number of bits of data, while that of the direct DLCT is N times the number of bits of data for calculating a spectra.

When the window slides continuously, the Sp-DLCT calculates the spectra of data for each window separately. To fully utilize the similarity among blocks of samples, the SDLCT is derived and shown to be more effective than classical methods. When the window continuously slides a large number of times, the SDLCT is also faster than the Sp-DLCT. The simulations showed that the SDLCT can realize the selectivity of signal spectra outputs. The method is used to compute a single frequency spectral bin, all N spectra or the desired frequency range according to the given application requirements.

The properties of the Sp-DLCT and SDLCT associated with additivity and reversibility were discussed in detail. The simulation results showed that the additivity property of the SDLCT can be achieved with an error as small as that of the Sp-DLCT. In addition, the Sp-DLCT has perfect reversibility, which is proved by numerical examples. With the reversibility property, the inverse transform of the proposed Sp-DLCT and SDLCT can be realized by the forward Sp-DLCT and SDLCT, respectively. To reduce spectral leakage, we used the discrete convolution theorem to perform time-domain windowing in the

frequency domain. The calculation of the DLCTs of data with a non-rectangular window can be translated into SDLCTs. This method guarantees the simplicity of the SDLCT.

APPENDIX A PROOF OF THEOREM 1

It is shown in [20] and [22] that the DLCT has the time-shift, frequency-shift and chirp periodic properties. The shifting theorems allow us to efficiently convert the algorithm into a zero-centered algorithm. Thus, the derivation of the general SDLCT can be understood starting with the definition of the DLCT of a contiguous subset of a longer sequence. For this derivation, we consider a DLCT of length N calculated on a subset of an input sequence $\tilde{N}$, ($N \leq \tilde{N}$). We start the derivation with the definition of the m th DLCT [28] bin for the $(n-1)$ th time window,

$$X_{n-1}^A(m) = C \sum_{k=-N/2}^{N/2-1} K_A(m, k) x \left[n-1 - \left(\frac{N}{2} - 1 \right) + k \right]. \quad (19)$$

Then, the DLCT of the n th time window sequence is

$$X_n^A(m) = C \sum_{k=-N/2}^{N/2-1} K_A(m, k) x \left[n - \left(\frac{N}{2} - 1 \right) + k \right]. \quad (20)$$

Substituting $k = k + 1$ into (20), we have

$$X_n^A(m) = C \sum_{k=-N/2+1}^{N/2} K_A(m, k-1) x \left[n-1 - \left(\frac{N}{2} - 1 \right) + k \right], \quad (21)$$

where

$$K_A(m, k-1) = e^{j\pi \frac{\alpha}{N|\beta|} (2\beta m + \gamma)} e^{-j\pi \frac{\alpha}{N|\beta|} [2m \frac{\gamma}{\beta} + (\frac{\gamma}{\beta})^2]} K_A \left(m + \frac{\gamma}{\beta}, k \right). \quad (22)$$

Combining (22) and (21), we have

$$X_n^A(m) = C e^{-j\pi \frac{\alpha}{N|\beta|} (\frac{\gamma^2}{\beta^2} + 2m \frac{\gamma}{\beta})} e^{j\pi \frac{1}{N} (2\beta m + \gamma)} \times \sum_{k=-N/2+1}^{N/2} x \left[n-1 - \left(\frac{N}{2} - 1 \right) + k \right] K_A \left(m + \frac{\gamma}{\beta}, k \right). \quad (23)$$

Next, we change the summation by separating the N th component and adding the $k = -N/2$ case to the summation, then,

subtracting it formally,

$$\begin{aligned}
 X_n^A(m) &= e^{j\pi \frac{1}{N|\beta|} \left[-\alpha(2m\frac{\gamma}{\beta} + \frac{\gamma^2}{\beta^2}) + 2m\beta + \gamma \right]} \\
 &\times \sum_{k=-N/2}^{N/2-1} K_A \left(m + \frac{\gamma}{\beta}, k \right) x \left[n-1 - \left(\frac{N}{2} - 1 \right) + k \right] \\
 &+ Cx[n]K_A \left(m, \frac{N}{2} - 1 \right) - Cx[n-N]K_A \left(m, \frac{-N}{2} - 1 \right). \quad (24)
 \end{aligned}$$

Note that the summation is the DLCT of the $(n-1)$ th time-window data, but equation (24) is the DLCT of the n th window. Therefore, the sliding DLCT can be expressed as

$$\begin{aligned}
 X_n^A(m) &= e^{j\pi \frac{1}{N|\beta|} \left[-\alpha(2m\frac{\gamma}{\beta} + \frac{\gamma^2}{\beta^2}) + 2m\beta + \gamma \right]} X_{n-1}^A \left(m + \frac{\gamma}{\beta} \right) \\
 &- Cx(n-N)K_A \left(m, \frac{-N}{2} - 1 \right) + Cx(n)K_A \left(m, \frac{N}{2} - 1 \right). \quad (25)
 \end{aligned}$$

Note that m is an integer in (25) denoting the m th LCT spectra. Similarly, $m + \gamma/\beta$ is the $(m + \gamma/\beta)$ th LCT spectra; this then requires that the value of γ/β be an integer. We can utilize the chirp periodic property and shift theorem of $X_{n-1}^A(m)$ in terms of (25) to obtain $X_n^A(m)$.

When γ/β is not an integer, we obtain the following formula associated with the modulation property of the sequence:

$$X_{n-1}^A \left(m + \frac{\gamma}{\beta} \right) = e^{j\pi \frac{1}{N|\beta|} \left[\alpha(2m\frac{\gamma}{\beta} + \frac{\gamma^2}{\beta^2}) \right]} G_{n-1}^A(m), \quad (26)$$

where $G_n^A(m)$ is the m th DLCT of the $(n-1)$ th time window of the signal $g(n-1, k) = x[n-1 - (N/2-1) + k]e^{-j\pi \frac{2\gamma k}{N|\beta|}}$. Substituting (26) into (25), we have

$$X_n^A(m) = Q \left[G_{n-1}^A(m) - Cx(n-N)L + Cx(n)P \right]. \quad (27)$$

Theorem 1 has been proven.

APPENDIX B PROOF OF THEOREM 2

Proof: The DFTs of (15) can be expressed as

$$\begin{aligned}
 W(m) &= \frac{1}{N} \sum_{k=-N/2}^{N/2-1} w(k) e^{-j\pi \frac{2\pi k m}{N}} \\
 &= \frac{1}{2N} \sum_{l=0}^{K-1} (-1)^l \alpha_l \sum_{k=-N/2}^{N/2-1} \left(e^{-j\pi \frac{2\pi k(m-l)}{N}} + e^{-j\pi \frac{2\pi k(m+l)}{N}} \right). \quad (28)
 \end{aligned}$$

It can be observed that (28) contains only a few non-zero samples; then, according to (14) and finite-point convolution, we can easily obtain Theorem 2. ■

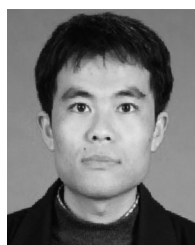
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